



COURSE SYLLABUS						
Course Code	IT INTE-E4	Course Title	IT Elective 4 – .NET Fundamentals		Course Credit	3
Course Description		This course provides students with foundational knowledge and practical skills in developing .NET applications using C#. It covers object-oriented programming, database connectivity, web development with ASP.NET, and cloud deployment through Microsoft Azure, emphasizing hands-on implementation and industry’s best practices.				
Pre-Requisites			Co-Requisites	None		
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		Advance an inclusive, equitable, and globally relevant polytechnic education towards national development.				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				



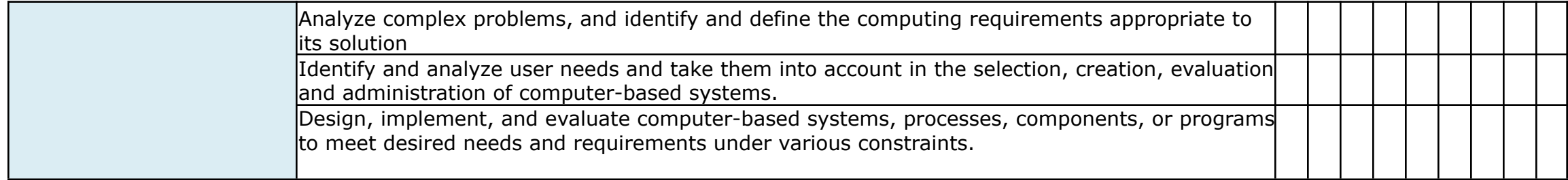
**INSTITUTIONAL
LEARNING OUTCOMES
(ILO)**

- As a polytechnic state university, PUP shall develop its students to possess:
1. **Critical and Creative Thinking.** Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world.
 2. **Effective Communication.** Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances.
 3. **Strong Service Orientation.** Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors.
 4. **Adept and Responsible Use or Development of Technology.** Graduates demonstrate optimized and responsible use of state- of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills.
 5. **Passion for Lifelong Learning.** Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.



	6. Leadership and Organizational Skills. Graduates assume leadership roles and become leading professionals in their respective disciplines by equipping them with appropriate organizational skills. 7. Personal and Professional Ethics. Graduates manifest integrity and adherence to moral and ethical principles in their personal and professional circumstances. 8. Resilience and Agility. Graduates demonstrate flexibility and the growth mindset to adapt and thrive in the volatile, uncertain, complex and ambiguous (VUCA) environment. 9. National and Global Responsiveness. Graduates exhibit a deep sense of nationalism as it complements the need to live as part of the global community where diversity is respected. They promote and fulfill various advocacies for human and social development.
COLLEGE GOALS AND OBJECTIVES	The Polytechnic University of the Philippines – Taguig Campus aims to achieve the following goals: 1. Offer curricula which are relevant and responsive to the changing needs of the industry and society; 2. Provide trainings to students that will enable them to become potent instruments to socio-economic development. 3. Prepare students to become productive, creative, innovative, and dynamic in their chosen fields of specialization. 4. Provide state-of-the-arts facilities of learning to optimize student development; 5. Inculcate appropriate values that are necessary to build a humane, disciplined, nationalist, and independent society; 6. Tap potentials of students, faculty, administrative staff and other stakeholders in formulating policies for institutional development; 7. Involve all stakeholders in enhancing programs for student’s faculty and employees’ development; 8. Partner with industry and other research institutions strengthening research capabilities of faculty, employees and students; 9. Facilitate presentation of research outputs in international fora, and their publication in recognized journals; 10.Undertake outreach and research-based extension programs by tapping the expertise and other resources of all stakeholders and, 11.Produce technologies for commercialization of livelihood improvement.

PROGRAM LEARNING OUTCOMES (PLO)	Based on CMO	Alignment to ILOs								
		1	2	3	4	5	6	7	8	9



THE COUNTRY'S 1st POLYTECHNICU





	Integrate IT-based solutions into the user environment effectively.									
	Assist in the creation of an effective IT project plan.									
	Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession									
	Understand best practices and standards and their applications.									
	Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.									
	Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.									
	Analyze the local and global impact of computing information technology on individuals, organizations, and society.									
	Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.									
	Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development									

COURSE LEARNING OUTCOMES (CLO)		Alignment to PLOs							
		1	2	3	4	5	6	7	8
	Explain the fundamental concepts, architecture, and components of the Microsoft .NET framework.	X						X	
	Apply C# programming constructs and object-oriented principles in developing structured and modular applications.	X	X		X			X	
	Implement data management and connectivity using ADO.NET and Entity Framework.		X	X	X			X	



	Develop interactive web applications using ASP.NET technologies.			X	X	X		X	
	Integrate web and cloud-based functionalities through Microsoft Azure services.				X	X		X	
	Demonstrate the use of APIs, authentication, and session management for secure and efficient systems.		X	X	X				X
	Utilize debugging, logging, and version control tools to maintain and optimize .NET applications.				X			X	X
	Design, build, and deploy a complete .NET project that showcases problem-solving, technical proficiency, and adherence to software development best practices.	X	X	X	X	X	X	X	



OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Desired Learning Outcomes (DLOs)	Learning Content/Topics	Instructional Delivery Design		Assessment		Alignment to CLOs			
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)			1	2	3	4
							Face-to-Face	FLTAs		
1	- Equip students on the pivotal ideas embedded in the vision, mission, and 10 Pillars of PUP - Gain an overview of the course subject and requirements	- Orientation about PUP Vision, Mission, and 10 Pillars - Overview about the Subject Course and Course Requirements	Present the VMGO, policies, and guidelines on the conduct of Face-to-Face; and of the requirements of the course.		Formative Assessment (Sharing of Learning)		X			
2	- Students will understand course goals, grading, tools, and .NET ecosystem. - Demonstrate basic C# syntax and program structure.	- Course orientation; .NET overview (CLR, .NET Core vs Framework); Visual Studio/VS Code setup. - C# program structure, data types, variables, operators, expressions.	Syllabus walkthrough; live IDE installation and hands-on setup. Live coding examples; short in-class exercises.	Recorded setup videos; step-by-step installation guide; reading assignment. Short video demos; practice problems on LMS.	Participation in orientation; quick IDE checklist (in-class). Short in-class coding quiz.	Submission of screenshot of working IDE; online checklist. Online coding exercises	X			



						(auto-graded) submission.				
3	Use control flow and methods to build modular code.	Control statements (if/switch/loops), functions/methods, scope.	Problem solving lab in class.	Guided coding lab files; pair programming via discussion board.	In-class problem solving exercise.	Submit functions collection and short reflection.	X	X		
4	Apply object-oriented programming (OOP) principles in C#.	Classes/objects, constructors, fields/properties, methods.	Lecture + OOP coding workshop.	Reading/case study on OOP design; recorded examples.	OOP mini-lab (create classes) in class.	OOP assignment (extend classes) submission.	X	X		
5	Implement inheritance, polymorphism, and abstraction.	Inheritance, virtual/override, interfaces, abstract classes.	Live demonstration of inheritance patterns; lab.	Design exercises; discussion board design critique.	Lab: implement class hierarchy (in class).	Assignment: refactor code to use interfaces /abstraction.	X	X		
6	Handle exceptions and perform file I/O reliably.	Exception handling, custom exceptions, file read/write, streams.	Code walkthrough + debugging session.	File I/O practice tasks; debugging videos.	In-class debugging exercise.	Submit file I/O mini-project and log.	X	X	X	
7	Use collections and generics to manage data.	Arrays, List<T>, Dictionary<TKey, TValue>, Queue, Stack, LINQ intro.	Hands-on lab using collections; LINQ demo.	Practice problems; sample datasets for	In-class mini quiz & lab.	Online coding tasks using	X	X		



				offline work.		collections				
--	--	--	--	---------------	--	-------------	--	--	--	--

WORLD UNIVERSITY RANKINGS

2025

7th in the Philippines
92nd in Southeast Asia
541-560 in Asia

WURI

The WORLD UNIVERSITY RANKINGS for INNOVATION

Rank 21
GENERATIVE AI

Rank 39
STUDENT SUPPORT

Rank 50
SYMBOL/

QS STARS

★★★★★

★★★★★

★★★★★

Employability

Teaching

Online Learning

Academic Development

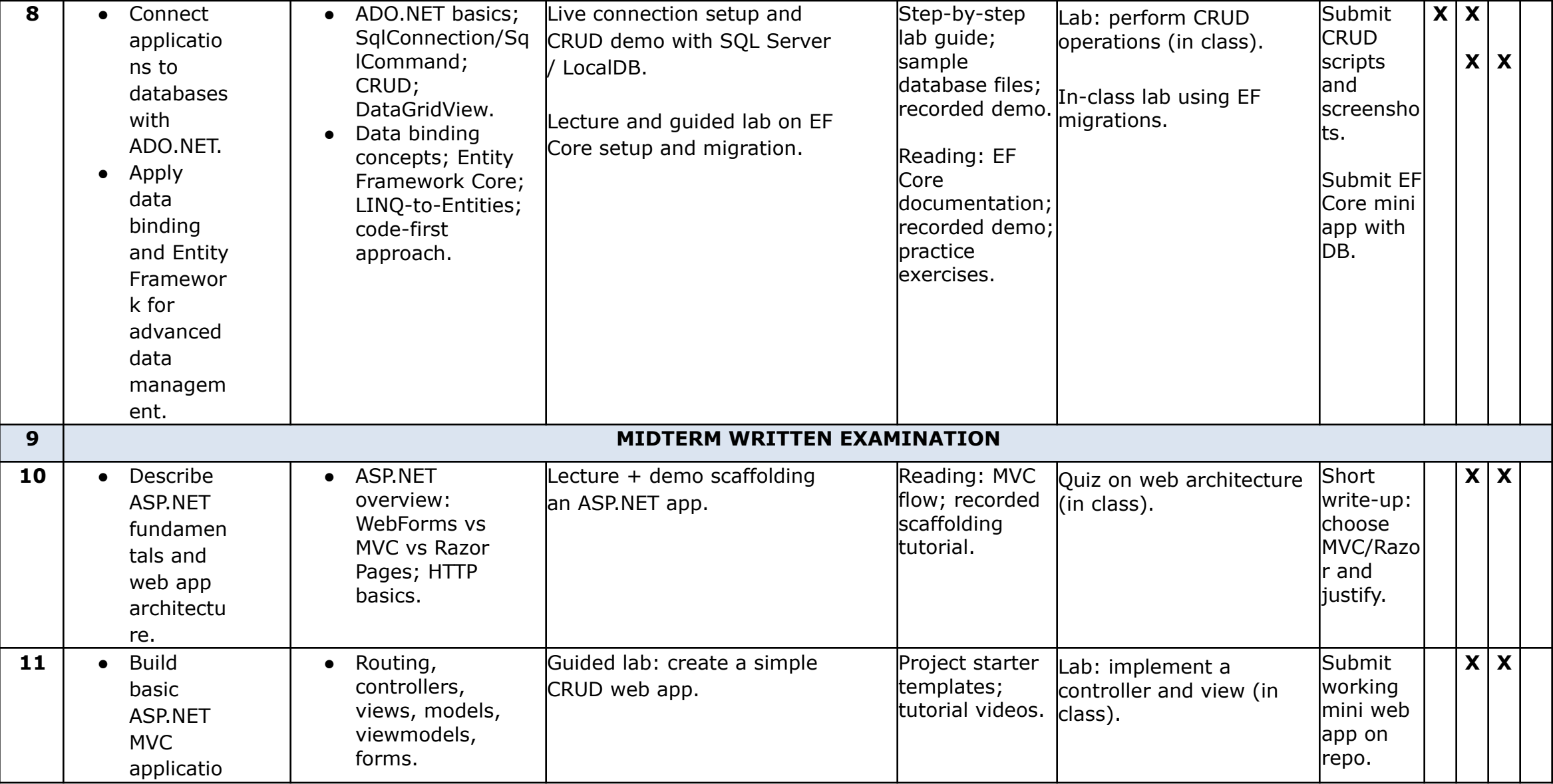
Internationalization

Social Responsibility

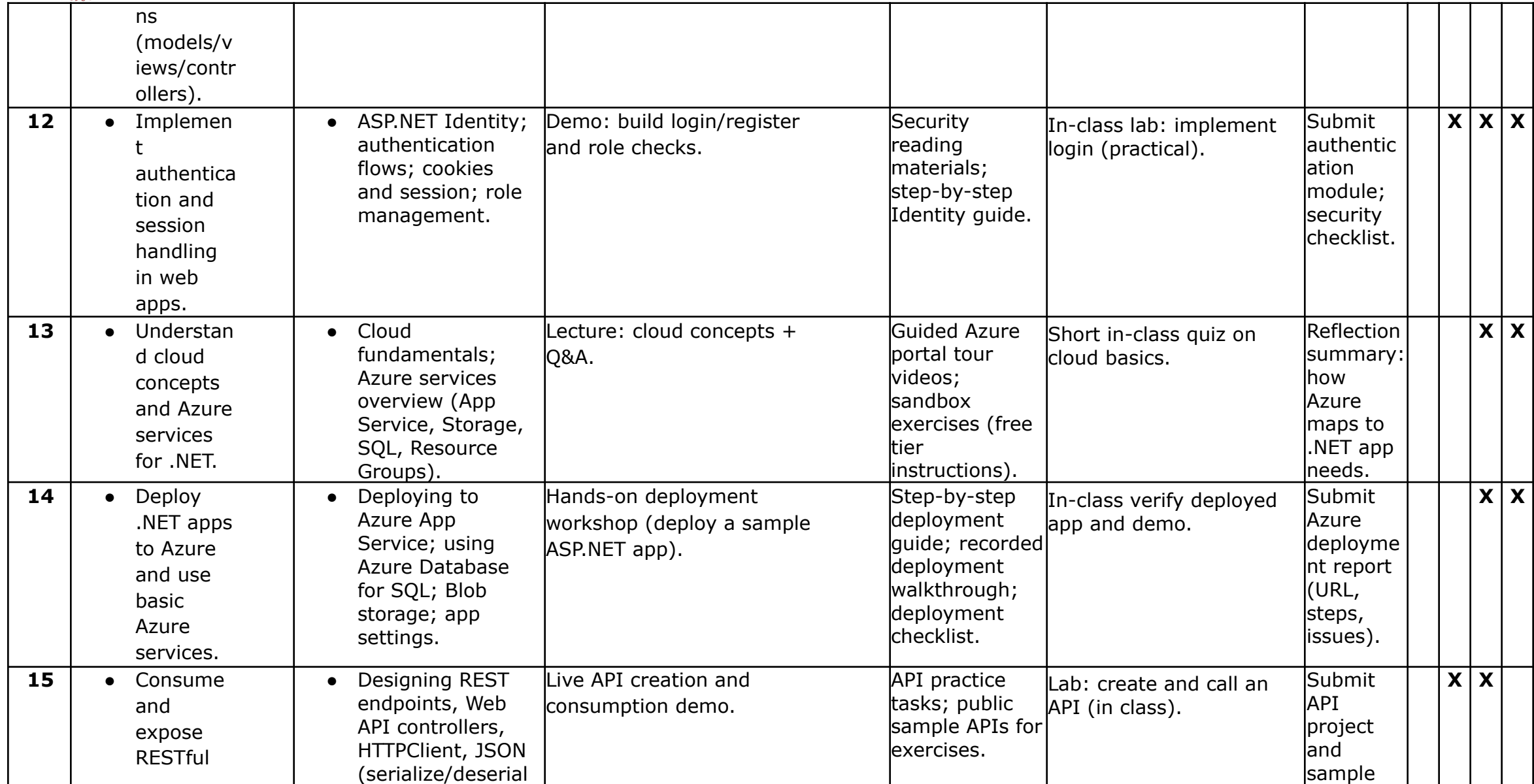
Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)

Times Higher Education Impact Rankings 2024

DEBUTS AT 1501+

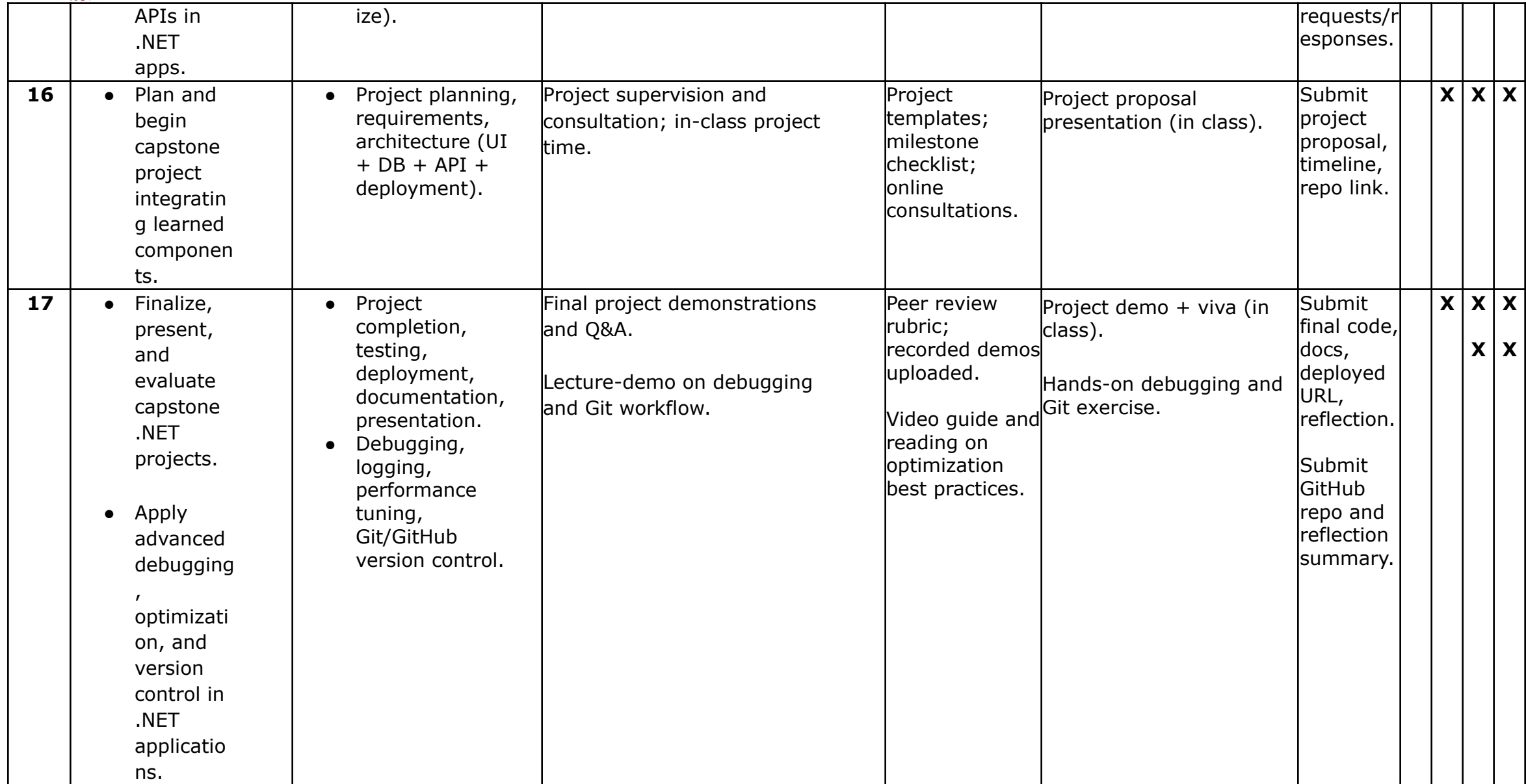


General Santos Avenue, Lower Bicutan, Taguig City, Philippines 1632
Direct Line: (02) 8837 5858 to 60 | Email: taguig@pup.edu.ph
Website: www.pup.edu.ph | Inquiries: <https://bit.ly/PUPSINTA>



THE COUNTRY'S 1st POLYTECHNICU





THE COUNTRY'S 1st POLYTECHNICU





18	FINAL WRITTEN EXAMINATION

General Santos Avenue, Lower Bicutan, Taguig City, Philippines 1632
Direct Line: (02) 8837 5858 to 60 | Email: taguig@pup.edu.ph
Website: www.pup.edu.ph | Inquiries: <https://bit.ly/PUPSINTA>

THE COUNTRY'S 1st POLYTECHNICU

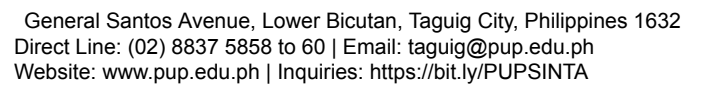




		Resources and Applying Strategies in Teaching Prose <i>Competency 4:</i> Recognizing Assessment in Teaching Prose	achievement which are consistent with the selected competencies ● Learning Plan Making according to the English curricula that develops higher order thinking skills of learners through the use of prose and dramatic texts		● Quizzes					
18	FINAL EXAMINATION									
LIST OF REFERENCES BOOKS: WEBSITE: Albahari, J., & Albahari, B. (2024). <i>C# 12 in a Nutshell: The definitive reference</i> . O’Reilly Media. Esposito, D., & Willis, A. (2023). <i>Modern web development with ASP.NET Core 8: Building scalable web applications</i> . Microsoft Press. Freeman, A. (2024). <i>Pro ASP.NET Core 8: Develop cloud-ready web applications using MVC, Blazor, and Razor Pages</i> . Apress. Liberty, J., & MacDonald, M. (2023). <i>Programming C# 12: Build cloud, web, and desktop applications</i> . O’Reilly Media. Microsoft. (2025). <i>Microsoft Learn: .NET documentation</i> . Microsoft Docs. https://learn.microsoft.com/dotnet Price, M. (2023). <i>C# 12 and .NET 8 – Modern cross-platform development</i> . Packt Publishing. Roth, D., & Hariri, S. (2024). <i>Blazor WebAssembly by Example: Build interactive web UIs using C# and .NET 8</i> . Packt Publishing.										



CLASSROOM POLICY	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)
Please refer to the FLTA policies, except for the stipulation on Attendance. Attendance will be checked during face-to-face sessions.	Students' Responsibilities Students are expected to: a. Take time to read and watch the provided class learning materials. b. Fulfill class requirements such as posting on discussion boards and submitting group presentations. c. Check the status of their submissions and inform the instructor if they are having problems submitting the requirements.



THE COUNTRY'S 1st POLYTECHNICU





The instructor may not be able to remind all students about their missing/unsubmitted requirements. If at the end of the course students still have missing requirements, the instructor will compute for the final grade without those submissions (They will be graded with 0) and submit the grade. If the final grade is "INC" grade, the student must contact the instructor to decide in completing the course, ASAP.

d. Confer with classmates or the instructor about the discussions during the synchronous sessions that they missed if unable to attend the live sessions. Recordings will be provided but may not be available immediately.

e. Inform the instructor of concerns that may prevent them from fully participating in the class. This includes withdrawal from the course, long term illnesses, personal matters, and others. Teachers will not be able to help you if you don't tell them of these concerns.

Attendance.
We recognize that students have varying access levels to technology and connectivity, which is why for the most part, the course is conducted asynchronously. During synchronous sessions, there will be no checking of attendance. However, everyone is encouraged to participate in these sessions. A recording and/or transcript of synchronous classes will be made available for students who are unable to attend a synchronous session.

Late Submissions.
Deadlines will be set for the submission of course requirements. This allows the instructor sufficient time to review them. It also helps students pace their work. If you experience problems submitting your work on time, please contact the instructor ahead of time. All requirements must be completed to pass the course. Late submissions will not merit deductions but may not receive timely feedback.

Cheating and Plagiarism.
Any work submitted by a student in this course for academic credit will be the student's own work. Any form of academic dishonesty will be fully dealt with according to the stipulations of the PUP Student Handbook.

Participation.
Participation during discussions is greatly encouraged, and students should feel free to ask questions when an idea or lesson is unclear or confusing, especially during synchronous sessions.

Announcements.
Class announcements will be communicated through the class president. Other communication shall always be coursed through university-issued (i.e., official) email accounts. Students must use their PUP student accounts.

Intellectual Property and Copyright.
Original materials created for this online class is covered by university policies on intellectual property. Some (third-party) instructional materials (e.g., YouTube videos) might be under the public domain or are open access, while others (e.g., some assigned readings) might be subject to fair use. All third-party instructional materials will



	be attributed to their original creators. Electronic copies of the assigned readings are for your private consumption only. Access to the course materials for this class for PUP students will be restricted to only those students registered in this course and will be available for only a
--	---



2025

7th in the Philippines
92nd in Southeast Asia
541-560 in Asia



The WORLD
UNIVERSITY
RANKINGS
for INNOVATION

Rank 21
GENERATIVE AI

Rank 39
STUDENT SUPPORT

Rank 50
SYMBOL/



★★★★★

★★★★★

★★★★★

Employability

Teaching

Online Learning

Academic Development

Inclusiveness

Internationalization

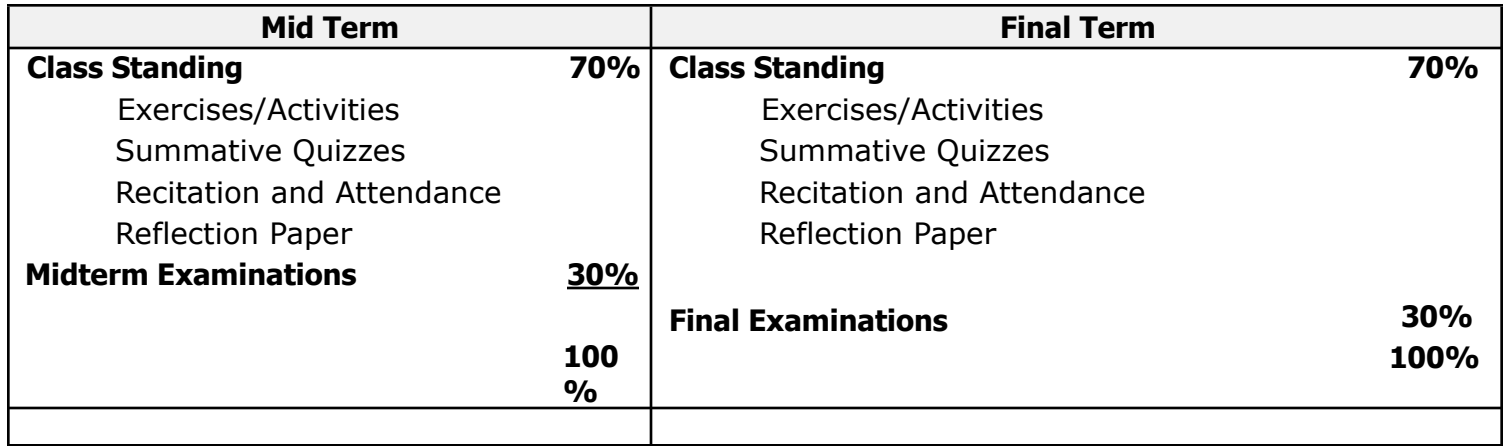
Social Responsibility

Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)



Times Higher Education
Impact Rankings 2024

DEBUTS AT 1501+



	limited period. Please do not share these materials with anyone elsewhere. The same applies to the recordings of the synchronous sessions.
COURSE REQUIREMENT/S	GRADING SYSTEM
Course requirements include active participation, completion of coding exercises and assignments, midterm and final exams, and a final .NET project with documentation and presentation. A short reflection or portfolio summarizing learning is also required.	$\text{Final Grade} = (\text{Midterm Grade} + \text{Final Term Grade}) / 2$

General Santos Avenue, Lower Bicutan, Taguig City, Philippines 1632
Direct Line: (02) 8837 5858 to 60 | Email: taguig@pup.edu.ph
Website: www.pup.edu.ph | Inquiries: <https://bit.ly/PUPSINTA>

THE COUNTRY'S 1st POLYTECHNICU



CLASS INFORMATION	FACULTY INFORMATION
Course, Year and Section: BSIT-TG-4-1	Name of Faculty: Manlapaz, Mark B.
Day and Time: TH – 07:30 AM – 12:30 PM	Consultation Time:
Room: TGA205	Office Tel. No./ Mobile Phone No.
School Year: 2025-2026 Semester: First	E-Mail Address:

Prepared by:	Reviewed and checked by:	Recommending Approval:	Approved by:
Faculty Member	MS. DIANNE S. SEGURORA Head of Academic Programs	DR. RHYAN V. MOLINAR Campus Director	DR. ARMANDO A. TORRES Vice President for Campuses